

Math 550, S21
Exam 1

Name: *Answer*

Instructions:

- There are totally 5 problems.
- You must show all of your work to receive a credit for a correct answer.
- No calculator is allowed in the exam.

Problem	Points	Score
1	20	
2	22	
3	20	
4	22	
5	20	
Total	104	

Good Luck !

Problem 1. (20 points) Given two vectors $\mathbf{a} = \sqrt{3}\mathbf{i} + 3\mathbf{j} + 2\sqrt{6}\mathbf{k}$ and $\mathbf{b} = (-\sqrt{3}, 0, \sqrt{6})$.
 (a) Find the angle θ between $\mathbf{a}$ and $\mathbf{b}$.

$$\|\mathbf{a}\| = \sqrt{(\sqrt{3})^2 + 3^2 + (2\sqrt{6})^2} = \sqrt{36} = 6$$

$$\|\mathbf{b}\| = \sqrt{(-\sqrt{3})^2 + 0^2 + (\sqrt{6})^2} = \sqrt{9} = 3$$

$$\begin{aligned} \mathbf{a} \cdot \mathbf{b} &= (\sqrt{3}, 3, 2\sqrt{6}) \cdot (-\sqrt{3}, 0, \sqrt{6}) \\ &= \sqrt{3} \cdot (-\sqrt{3}) + 3 \cdot 0 + 2\sqrt{6} \cdot \sqrt{6} = 9 \end{aligned}$$

$$\cos \theta = \frac{\mathbf{a} \cdot \mathbf{b}}{\|\mathbf{a}\| \|\mathbf{b}\|} = \frac{9}{3 \cdot 6} = \frac{1}{2}$$

$$\Rightarrow \theta = \arccos\left(\frac{1}{2}\right) = \frac{\pi}{3}$$

(b) Compute $\mathbf{a} \times \mathbf{b}$ and use it to find the area of the parallelogram determined by $\mathbf{a}$ and $\mathbf{b}$.

$$\mathbf{a} \times \mathbf{b} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \sqrt{3} & 3 & 2\sqrt{6} \\ -\sqrt{3} & 0 & \sqrt{6} \end{vmatrix}$$

$$= \vec{i} \begin{vmatrix} 3 & 2\sqrt{6} \\ 0 & \sqrt{6} \end{vmatrix} - \vec{j} \begin{vmatrix} \sqrt{3} & 2\sqrt{6} \\ -\sqrt{3} & \sqrt{6} \end{vmatrix} + \vec{k} \begin{vmatrix} \sqrt{3} & 3 \\ -\sqrt{3} & 0 \end{vmatrix}$$

$$= 3\sqrt{6}\vec{i} - 9\sqrt{2}\vec{j} + 3\sqrt{3}\vec{k}$$

$$\text{Area} = \|\mathbf{a} \times \mathbf{b}\| = \sqrt{(3\sqrt{6})^2 + (9\sqrt{2})^2 + (3\sqrt{3})^2}$$

$$= \sqrt{243} = 9\sqrt{3}$$

Problem 2. (22 points) Given three points $P(-1, 3, 1)$, $Q(0, 5, -2)$, and $R(4, 3, -1)$ in $\mathbb{R}^3$.

(a) Let $\mathbf{L}$ be the line through the two points P and Q . Find parametric equations for $\mathbf{L}$.

$$\vec{PQ} = (0, 5, -2) - (-1, 3, 1) = (1, 2, -3)$$

$$\Rightarrow \mathbf{r}(t) = (-1, 3, 1) + t(1, 2, -3), \quad -\infty \leq t \leq +\infty$$

$$\text{i.e. } \begin{cases} x(t) = -1 + t, \\ y(t) = 3 + 2t, \\ z(t) = 1 - 3t, \end{cases} \quad -\infty \leq t \leq +\infty$$

(b) Find an equation for the plane that contains the triangle $\triangle PQR$.

$$\vec{PQ} = (1, 2, -3), \quad \vec{PR} = (4, 3, -1) - (-1, 3, 1) = (5, 0, -2)$$

$$\Rightarrow \vec{n} = \vec{PQ} \times \vec{PR} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 1 & 2 & -3 \\ 5 & 0 & -2 \end{vmatrix}$$

$$= \vec{i} \begin{vmatrix} 2 & -3 \\ 0 & -2 \end{vmatrix} - \vec{j} \begin{vmatrix} 1 & -3 \\ 5 & -2 \end{vmatrix} + \vec{k} \begin{vmatrix} 1 & 2 \\ 5 & 0 \end{vmatrix} = -4\vec{i} - 13\vec{j} - 10\vec{k}$$

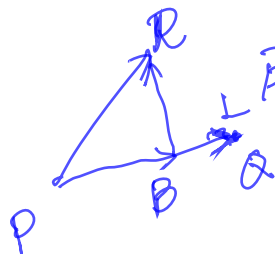
Thus we have the equation of the plane as

$$-4(x+1) - 13(y-3) - 10(z-1) = 0$$

$$4x + 13y + 10z = 45$$

$$\frac{\sqrt{429}}{\sqrt{14}}$$

(c) Find the distance between the point R and the line $\mathbf{L}$ defined in (a).



$$\vec{PR} = (5, 0, -2), \quad \vec{PQ} = (1, 2, -3)$$

$$\vec{PB} = \text{Proj}_{\vec{PQ}} \vec{PR} = \left(\frac{\vec{PR} \cdot \vec{PQ}}{\vec{PQ} \cdot \vec{PQ}} \right) \vec{PQ} = \frac{(5, 0, -2) \cdot (1, 2, -3)}{(1, 2, -3) \cdot (1, 2, -3)} \cdot (1, 2, -3)$$

$$= \frac{11}{14} (1, 2, -3)$$

$$\begin{aligned} \text{Dist} &= \|\vec{BR}\| = \|\vec{PR} - \vec{PB}\| \\ &= \|(5, 0, -2) - \frac{11}{14}(1, 2, -3)\| = \frac{1}{14} \|(59, -22, 5)\| \\ &= \frac{1}{14} \sqrt{59^2 + (-22)^2 + 5^2} = \frac{1}{14} \sqrt{3990} \end{aligned}$$

Problem 3. (20 points)

(a) Let $z = f(x, y)$ be any scalar function with continuous partial derivatives, and define $x = e^r \cos \theta$ and $y = e^r \sin \theta$. Use the **Chain Rule** to show

$$\left(\frac{\partial z}{\partial x}\right)^2 + \left(\frac{\partial z}{\partial y}\right)^2 = e^{-2r} \left[\left(\frac{\partial z}{\partial r}\right)^2 + \left(\frac{\partial z}{\partial \theta}\right)^2 \right].$$

Proof: $\frac{\partial z}{\partial r} = \frac{\partial z}{\partial x} \frac{\partial x}{\partial r} + \frac{\partial z}{\partial y} \frac{\partial y}{\partial r} = \frac{\partial z}{\partial x} e^r \cos \theta + \frac{\partial z}{\partial y} e^r \sin \theta$

$$\Rightarrow \left(\frac{\partial z}{\partial r}\right)^2 = e^{2r} \left[\left(\frac{\partial z}{\partial x}\right)^2 \cos^2 \theta + \left(\frac{\partial z}{\partial y}\right)^2 \sin^2 \theta + 2 \frac{\partial z}{\partial x} \frac{\partial z}{\partial y} \cos \theta \sin \theta \right]$$

$$\frac{\partial z}{\partial \theta} = \frac{\partial z}{\partial x} \frac{\partial x}{\partial \theta} + \frac{\partial z}{\partial y} \frac{\partial y}{\partial \theta} = \frac{\partial z}{\partial x} (-e^r \sin \theta) + \frac{\partial z}{\partial y} (e^r \cos \theta)$$

$$\Rightarrow \left(\frac{\partial z}{\partial \theta}\right)^2 = e^{2r} \left[\left(\frac{\partial z}{\partial x}\right)^2 \sin^2 \theta + \left(\frac{\partial z}{\partial y}\right)^2 \cos^2 \theta - 2 \frac{\partial z}{\partial x} \frac{\partial z}{\partial y} \sin \theta \cos \theta \right]$$

Thus, $e^{-2r} \left[\left(\frac{\partial z}{\partial r}\right)^2 + \left(\frac{\partial z}{\partial \theta}\right)^2 \right]$

$$= e^{-2r} \cdot e^{2r} \left[\left(\frac{\partial z}{\partial x}\right)^2 (\cos^2 \theta + \sin^2 \theta) + \left(\frac{\partial z}{\partial y}\right)^2 (\sin^2 \theta + \cos^2 \theta) \right]$$

$$= \left(\frac{\partial z}{\partial x}\right)^2 + \left(\frac{\partial z}{\partial y}\right)^2$$

(b) Find an equation for the tangent plane to the paraboloid $x^2 + \frac{y^2}{4} + \frac{z^2}{9} = 3$ at the point $(1, 2, 3)$.

$$f(x, y, z) = x^2 + \frac{y^2}{4} + \frac{z^2}{9} - 3$$

$$\nabla f(x, y, z) = (2x, \frac{1}{2}y, \frac{2}{9}z)$$

$$\Rightarrow \nabla f(1, 2, 3) = (2, 1, \frac{2}{3}) \rightarrow \vec{n} = \text{the normal of the plane.}$$

The tangent plane at $(1, 2, 3)$ is then given by

$$2(x-1) + (y-2) + \frac{2}{3}(z-3) = 0$$

$$2x + y + \frac{2}{3}z = 6$$

Problem 4. (22 points) Let $\mathbf{x}(t) = (e^t \sin t, 1, e^t \cos t)$ that defines a path in $\mathbb{R}^3$.

(a) Find the unit tangent vector $\mathbf{T}$ of the path.

$$\begin{aligned} \mathbf{x}'(t) &= (e^t \sin t + e^t \cos t, 0, e^t \cos t - e^t \sin t) \\ &= (e^t (\sin t + \cos t), 0, e^t (\cos t - \sin t)) \\ \Rightarrow \|\mathbf{x}'(t)\| &= \sqrt{e^{2t} (\sin t + \cos t)^2 + 0^2 + e^{2t} (\cos t - \sin t)^2} \\ &= \sqrt{e^{2t} (1 + 2 \sin t \cos t) + e^{2t} (1 - 2 \cos t \sin t)} \\ &= \sqrt{2e^{2t}} = \sqrt{2} e^t \\ \Rightarrow \mathbf{T}(t) &= \frac{\mathbf{x}'(t)}{\|\mathbf{x}'(t)\|} = \frac{1}{\sqrt{2}} (\sin t + \cos t, 0, \cos t - \sin t) \end{aligned}$$

(b) Find the length of the arc defined by the path with $0 \leq t \leq 1$.

$$\begin{aligned} L &= \int_0^1 \|\mathbf{x}'(t)\| dt = \int_0^1 \sqrt{2} e^t dt \\ &= \sqrt{2} e^t \Big|_{t=0}^{t=1} = \sqrt{2} (e^1 - e^0) = \sqrt{2} (e - 1) \end{aligned}$$

(c) Find the curvature κ and the principal unit normal vector $\mathbf{N}$ of the path.

$$\begin{aligned} \mathbf{T}'(t) &= \frac{1}{\sqrt{2}} (\cos t - \sin t, 0, -\sin t - \cos t) \\ \|\mathbf{T}'(t)\| &= \frac{1}{\sqrt{2}} \sqrt{(\cos t - \sin t)^2 + 0^2 + (-\sin t - \cos t)^2} \\ &= \frac{1}{\sqrt{2}} \sqrt{1 - 2 \cos t \sin t + 1 + 2 \sin t \cos t} \\ &= \frac{\sqrt{2}}{\sqrt{2}} = 1 \\ \Rightarrow \kappa(t) &= \frac{\|\mathbf{T}'(t)\|}{\|\mathbf{x}'(t)\|} = \frac{1}{\sqrt{2} e^t} = \frac{\sqrt{2}}{2} e^{-t} \\ \mathbf{N}(t) &= \frac{\mathbf{T}'(t)}{\|\mathbf{T}'(t)\|} = \frac{1}{\sqrt{2}} (\cos t - \sin t, 0, -\sin t - \cos t) \end{aligned}$$

Problem 5. (20 points)

(a) Compute the divergence and the curl of the vector field $\mathbf{F}(x, y, z) = y^2 z \mathbf{i} + e^{2xyz} \mathbf{j} + x^2 y \mathbf{k}$. Also please explain why $\mathbf{F}$ is not a gradient field.

$$\begin{aligned}\operatorname{div} \mathbf{F} &= \nabla \cdot \mathbf{F} = \frac{\partial}{\partial x}(y^2 z) + \frac{\partial}{\partial y}(e^{2xyz}) + \frac{\partial}{\partial z}(x^2 y) = 2xz e^{2xyz} \\ \operatorname{curl} \mathbf{F} &= \nabla \times \mathbf{F} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ y^2 z & e^{2xyz} & x^2 y \end{vmatrix} = \vec{i} \begin{vmatrix} \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ e^{2xyz} & x^2 y \end{vmatrix} - \vec{j} \begin{vmatrix} \frac{\partial}{\partial x} & \frac{\partial}{\partial z} \\ y^2 z & x^2 y \end{vmatrix} + \vec{k} \begin{vmatrix} \frac{\partial}{\partial x} & \frac{\partial}{\partial y} \\ y^2 z & e^{2xyz} \end{vmatrix} \\ &= \vec{i}(x^2 - 2xy e^{2xyz}) - \vec{j}(2xy - y^2) + \vec{k}(2yz e^{2xyz} - 2yz) \\ &= x(x - 2ye^{2xyz}) \vec{i} + y(y - 2x) \vec{j} + 2yz(e^{2xyz} - 1) \vec{k} \end{aligned}$$

$\vec{F}$ is not gradient field since it doesn't hold that

$$\nabla \times \mathbf{F} \equiv 0 \text{ for all } (x, y, z)$$

(b) Find the second order Taylor polynomial $P_2(x, y, z)$ for the function $f(x, y, z) = \frac{1}{x^2 + y^2 + z^2 + 2}$ at the point $\mathbf{a} = (0, 0, 0)$.

let $r = x^2 + y^2 + z^2 + 2$, then $f(x, y, z) = \frac{1}{r}$

$$\begin{aligned}Df &= \left[\frac{-2x}{r^2} \quad \frac{-2y}{r^2} \quad \frac{-2z}{r^2} \right] \Rightarrow \nabla f(0, 0, 0) = [0 \ 0 \ 0] \\ Hf &= \begin{bmatrix} -\frac{2}{r^2} + \frac{8x^2}{r^3} & \frac{8xy}{r^3} & \frac{8xz}{r^3} \\ \frac{8yx}{r^3} & -\frac{2}{r^2} + \frac{8y^2}{r^3} & \frac{8yz}{r^3} \\ \frac{8zx}{r^3} & \frac{8zy}{r^3} & -\frac{2}{r^2} + \frac{8z^2}{r^3} \end{bmatrix} = \begin{bmatrix} -\frac{1}{2} & 0 & 0 \\ 0 & -\frac{1}{2} & 0 \\ 0 & 0 & -\frac{1}{2} \end{bmatrix} \end{aligned}$$

thus

$$\begin{aligned}P_2(x, y, z) &= f(0, 0, 0) + Df(0, 0, 0) \begin{bmatrix} x \\ y \\ z \end{bmatrix} + \frac{1}{2} [x \ y \ z] Hf(0, 0, 0) \begin{bmatrix} x \\ y \\ z \end{bmatrix} \\ &= \frac{1}{2} + [0 \ 0 \ 0] \begin{bmatrix} x \\ y \\ z \end{bmatrix} + \frac{1}{2} [x \ y \ z] \begin{bmatrix} -\frac{1}{2} & 0 & 0 \\ 0 & -\frac{1}{2} & 0 \\ 0 & 0 & -\frac{1}{2} \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} \\ &= \frac{1}{2} - \frac{1}{4}x^2 - \frac{1}{4}y^2 - \frac{1}{4}z^2 \end{aligned}$$